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2.1 Data Representation
2.2 Statistical Data Analysis
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2.4 Dimensionality Reduction
2.5 Data Classification
2.6 Data Prediction

Video: Data Prediction
56 min

Graded App Item: EstimationErrorPlot
Submitted

Quiz: Predicting Fuel Efficiency Using Regression Trees
Submitted

Quiz: Predicting Fuel Efficiency Using Gaussian Process Regression
Submitted

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Predicting Fuel Efficiency Using Gaussian Process Regression

Review Learning Objectives

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Due Apr 21, 11:59 PM IST

Receive grade
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1. Load the 'carbig' dataset into MATLAB. Curate the data into a clean training and testing set using the following code: **1 / 1 point**

```
load carbig
D = [Acceleration, Cylinders, Displacement, Horsepower, Model_Year, Weight];
y = MPG;
msk = ~ismissing(y);
D = D(msk,:);
y = y(msk);


D_train  = D(1:2:end,:);
y_train = y(1:2:end);

D_test = D(2:2:end,:);
y_test = y(2:2:end);
rng(0);
```

Fit a Gaussian Process regression model to the training data D_train, y_train, and default parameters, with the exception of using the automatic hyperparameter optimization option. and the option:

```
'HyperparameterOptimizationOptions',struct('AcquisitionFunctionName','expected-improvement-plus')
```

which ensures the results are repeatable.

What is the name of the function we use to fit a Gaussian Process regression model?

fitrgp

Correct

2. Use the model with the predict function to estimate fuel efficiency for cars in the testing dataset (D_test). **1 / 1 point**

Compare the estimation with the true values (y_test). **What's the mean squared difference between the two, accurate to two decimal places?** You can use the EstimationErrorPlot function we made to find this.

6.9452

Correct

3. Which regression method performs better on this task? **1 / 1 point**

- ○ Regression trees
- ◉ Gaussian Process regression

Correct
Correct. With lower mean squared errors seen in the testing dataset, Gaussian Process regression appears to perform better on this test.

Advanced: This hypothesis could be tested using a statistical test (see paired t-tests with the 'ttest' function)